

The Body's Defence Army

II PUC · Ch 7 · Human Health and Disease

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



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Kingdom of Health

"**Health** is not just the absence of disease," said Prof. BAIO. "It is complete physical, mental and social well-being." Germs that cause disease are called **pathogens** — bacteria, viruses, fungi, protozoans and worms.

> 💡 **REMEMBER**
> **Diseases that spread from person to person are infectious diseases.**



The Invaders

Typhoid, caused by Salmonella typhi, spreads through dirty food and water and is confirmed by the **Widal test**. **Pneumonia** (Streptococcus, Haemophilus) fills the alveoli with fluid. The **common cold** is caused by rhinoviruses and affects the nose — not the lungs.

💡 **REMEMBER**

"Typhoid Mary" was a famous healthy carrier of typhoid.



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The Malaria Spy

A female Anopheles mosquito slipped **sporozoites** of **Plasmodium** into the blood. They multiplied in the liver, then attacked red blood cells. When the RBCs burst, a toxin called **haemozoin** caused chills and high fever every 3–4 days.

💡 REMEMBER

The sexual stage of Plasmodium happens inside the mosquito.



More Invaders

Amoebiasis (Entamoeba histolytica) is carried by houseflies. **Ascariasis** spreads through soil and water. **Filariasis** (Wuchereria), spread by Culex mosquitoes, causes elephantiasis. **Ringworm** is a fungal skin infection.

💡 **REMEMBER**

Prevention: clean food and water, personal hygiene, and controlling mosquitoes and flies.



Guards at the Gate

The first line of defence is **innate immunity**, present from birth. Physical barriers: skin and mucus. Physiological barriers: stomach acid, saliva, tears. Cellular barriers: neutrophils, monocytes, macrophages and NK cells. Infected cells even send **interferons** to protect their neighbours.

💡 REMEMBER

Innate immunity is non-specific.



The Special Forces

Acquired immunity remembers! **B-cells** make **antibodies** — Y-shaped H_2L_2 proteins (IgA, IgM, IgE, IgG). **T-cells** help them. Making your own antibodies, for example after a **vaccine**, is **active immunity**; getting ready-made antibodies, like IgA in colostrum, is **passive immunity**.

💡 **REMEMBER**

Primary lymphoid organs: bone marrow and thymus. Allergy is caused by histamine released from mast cells.



The Sneaky Virus

HIV, a retrovirus, first enters **macrophages** and turns them into "HIV factories". Then it attacks **helper T-cells**, and their number falls — the body loses its defence. This is **AIDS**. The **ELISA** test detects the infection.

💡 **REMEMBER**

HIV spreads through unsafe sex, infected blood, shared needles and from mother to child.



Cancer and Harmful Habits

Normal cells stop dividing when they touch others — **contact inhibition**. Cancer cells lose it. **Benign** tumours stay in place; **malignant** tumours spread by **metastasis**. **Carcinogens** include X-rays, UV rays and tobacco smoke. Drugs like heroin and cocaine, and tobacco's nicotine, harm the body — so say no!

💡 **REMEMBER**

Cancer treatment: surgery, radiation, chemotherapy and immunotherapy (α-interferon).



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

Health p.1 · pathogens p.1 · Typhoid p.2 · Widal test p.2 · Pneumonia p.2 · common cold p.2 · sporozoites p.3 · Plasmodium p.3 · haemozoin p.3 · Amoebiasis p.4 · Ascariasis p.4 · Filariasis p.4 · Ringworm p.4 · innate immunity p.5 · interferons p.5 · Acquired immunity p.6 · B-cells p.6 · antibodies p.6 · T-cells p.6 · vaccine p.6 · active immunity p.6 · passive immunity p.6 · HIV p.7 · macrophages p.7 · helper T-cells p.7 · AIDS p.7 · ELISA p.7 · contact inhibition p.8 · Benign p.8 · malignant p.8 · metastasis p.8 · Carcinogens p.8



QUICK QUIZ

Do you remember?

1. The Widal test confirms:

- Malaria
- Typhoid
- Pneumonia
- AIDS

2. Malaria fever is caused by the toxin:

- Histamine
- Haemozoin
- Interferon
- Serotonin

3. An antibody molecule is written as:

- H_2L_2
- H_4L_2
- HL
- H_2L_4

4. HIV attacks:

- Red blood cells
- Helper T-cells
- Platelets
- B-cells only

5. Spread of cancer cells to other parts is:

- Contact inhibition
- Metastasis
- Mitosis
- Immunity

Answers: 1-Typhoid · 2-Haemozoin · 3-H_2L_2 · 4-Helper T-cells · 5-Metastasis